

1_Compresión_del_problema

February 24, 2026

```
[1]: # Leemos información de documento Excel
import pandas as pd
dataFrame = pd.read_csv('../Material/Database.csv', sep=';')
print(dataFrame)
```

	ID	Class
0	OCTA1013661_19350130_1.png	Glaucoma
1	OCTA1153367_19320227_2.png	Glaucoma
2	OCTA1653454_19640601_2.png	Glaucoma
3	OCTADB5052_19500115_2.png	Glaucoma
4	OCTADB5551_19491106_1.png	Glaucoma
..	...	...
184	OCTADB7376_19590206_1.png	Healthy
185	OCTADB7376_19590206_2.png	Healthy
186	OCTADB7377_19620916_2.png	Healthy
187	OCTADB7398_19940401_1.png	Healthy
188	OCTADB7398_19940401_2.png	Healthy

[189 rows x 2 columns]

```
[2]: # Extraer información de interés
import numpy as np
clases, frec = np.unique(dataFrame.Class, return_counts=True)
print(clases, frec)
```

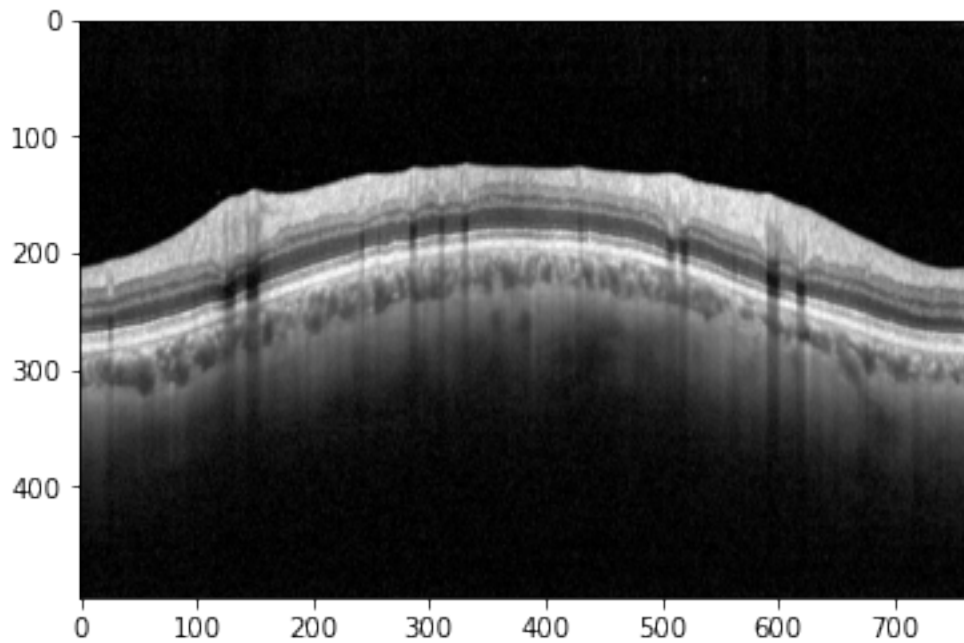
['Glaucoma' 'Healthy'] [93 96]

```
[10]: # Seleccionar una muestra aleatoria de nuestra base de datos
import random
import cv2
import matplotlib.pyplot as plt
num = random.randint(0, len(dataFrame))
print(num)
name_img = dataFrame.ID[num]
print(name_img)

img = cv2.imread('../Material/Images/' + name_img)
plt.imshow(img, cmap='gray')
plt.show()
```

142

OCTADB7271_19610921_1.png



```
[12]: # Leemos las máscaras y ploteamos los contornos sober la imagen original
rnfl_mask = cv2.imread('../Material/RNFL_masks/' + name_img, 0)
retina_mask = cv2.imread('../Material/Retina_masks/' + name_img, 0)

cont_rnfl, _ = cv2.findContours(rnfl_mask, cv2.RETR_TREE, cv2.
    ↳CHAIN_APPROX_SIMPLE)
cont_retina, _ = cv2.findContours(retina_mask, cv2.RETR_TREE, cv2.
    ↳CHAIN_APPROX_SIMPLE)

image = cv2.drawContours(img.copy(), cont_rnfl, -1, (255, 0, 0), 3)
image = cv2.drawContours(image.copy(), cont_retina, -1, (0, 255, 0), 3)

plt.imshow(image, cmap='gray')
plt.show()
```

